

RESEARCH ARTICLE

Does proactive boundary-spanning search drive green innovation? Exploring the significance of green dynamic capabilities and analytics capabilities

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Abstract

Successful innovation requires firms to search beyond their organizational confines for complementary knowledge resources ahead of their competitors. While such proactive boundary-spanning search (PBS) provides relevant knowledge resources for green innovations, it can also lead to information overload that derails green innovation efforts. Therefore, there is a need to understand how and when PBS translates into green innovation. This study combines the dynamic capabilities theory with the organizational information processing theory to develop a moderated mediation model that espouses the role of green dynamic capability and analytics capability in the relationship between PBS and green innovation. The study's hypothesis is tested using firm-level survey data from 200 manufacturing firms in Ghana. Partial least squares structural equations modeling (PLS-SEM) and PROCESS were used to evaluate the research model. The results confirm a positive relationship between PBS and green dynamic capabilities, and also between green dynamic capabilities and green innovation. More importantly, PBS drives green innovation through green dynamic capability, particularly at moderate and high levels of analytics capability. The theoretical and managerial implications of the study are outlined in the text.

KEYWORDS

analytics capability, Ghana, green dynamic capabilities, green innovation, proactive boundary-spanning search, sustainability

1 | INTRODUCTION

The environmental sustainability movement is on! Governments, businesses, and individuals are racing to achieve sustainable development goals (Appiah, 2019). Now more than ever, businesses are forced to heed the environmental demands of stakeholders (Appiah & Abul-Majeed, 2021; Yeşiltaş et al., 2022). One way firms can leverage resources to impact the natural environment and their firm positively is through green innovation (Gürlek & Koseoglu, 2021). Green innovation—the modification of products and processes to eliminate or reduce environmental impacts—creates harmony

between the environment, society, and the economy (Gürlek & Koseoglu, 2021; Lisi et al., 2020; Wang et al., 2019). Green innovation reduces environmental risks by emphasizing energy savings, pollution prevention, recycling, environmentally friendly products, and corporate environmental management (Arici & Uysal, 2022).

The growing importance of green innovation calls for an understanding of its determinants (Yang & Jiang, 2023). Successful green innovation requires knowledge resources that are seldom available in a single organization (Appiah, 2023). Therefore, firms must acquire and integrate knowledge from partners in the external environment—customers, suppliers, research institutions, and so forth—to support

their internal resources (Du et al., 2018; Kong et al., 2020; Lee & Kim, 2011; Martínez-Ros & Kunapatarawong, 2019; Yang & Lin, 2020). Resultantly, how firms search and integrate boundary-spanning knowledge resources affects their green innovation success (Stefan & Bengtsson, 2017; Yang et al., 2021). Proactive boundary-spanning search (PBS), which refers to the search for new and unrecognized knowledge ahead of competitors (Xue & Wang, 2023), enables firms to collect and use relevant knowledge resources from the external environment to support green innovation development. PBS enables firms to access novel and heterogeneous knowledge that facilitates successful green innovations (Yang et al., 2021). PBS also helps firms to access novel industry information that can facilitate “first-mover advantages” (Wang & Yang, 2023).

Scholars have yet to reach a consensus on how external knowledge search affects innovation (Zhu et al., 2018). Specifically, the relationship between PBS and green innovation is unclear (Xue & Wang, 2023), and has received scant research attention (Wang & Yang, 2023). PBS is a “double-edged sword” (Shi et al., 2019): while it facilitates access to external knowledge resources supporting green innovation (Yang et al., 2021; Ye et al., 2022), it comes with cost, integration challenges and the potential for information overload which can derail green innovation (Shi et al., 2019; Wang & Yang, 2023). An important question, therefore, is how and when does PBS result in green innovation? A few studies have contributed to resolving this question by examining how organizational resilience (Wang & Yang, 2023; Xue & Wang, 2023) and resource orchestration capabilities (Wang et al., 2019) mediate the relationship between PBS and green innovation. Despite these early advances, theoretical and empirical specification of the relationship between PBS and green innovation remains nascent (Wang & Yang, 2023).

The current volatile, uncertain, complex, and ambiguous (V.U.C.A) environment requires that beyond PBS, firms must develop capabilities to integrate and reconfigure environmental resources (Qiu et al., 2020). The focal firm's existing knowledge and capabilities determine the extent to which it can utilize knowledge from PBS (Wang et al., 2019). Green dynamic capability—a firm's capacity that enables timely and swift responses to stakeholder environmental demands—enables firms to utilize available knowledge resources to build new skills required for green innovation (Yousaf, 2021). Firms that lack green dynamic capabilities struggle to develop green innovations (Yousaf, 2021) while firms with strong green dynamic capabilities advertently implement green innovations (Achi et al., 2022). Although studies have shown that green dynamic capabilities positively affect green innovation (Yousaf, 2021), it is still unclear how firms can improve green dynamic capability through PBS to deal with the risks associated with green innovation. Drawing on the dynamics capability theory (DCT), this study examines the mediating role of green dynamic capabilities in the link between PBS and green innovation.

PBS is costly and time-consuming as early mover searches are associated with high risk and uncertainty (Yang et al., 2021). PBS can result in voluminous, complex, and highly unstructured data that

requires significant information processing (Katila & Chen, 2008). Organizational information processing theory (OIPT) suggests that firms can reduce task uncertainty by reducing information processing needs or increasing information processing capabilities (Galbraith, 1974; Yu et al., 2019). PBS increases the information processing needs of the focal firm, which requires firms to develop analytical capabilities that help to process “vast, various and valuable data quickly” (Liu & Wei, 2022). Thus, analytics capability—the ability to “process large volumes and varieties of data with the velocity required to gain relevant insights” (Liu & Wei, 2022)—may interact with PBS to enhance green innovation. While analytics is known to support green innovations (Zameer et al., 2020), how it conditions the outcomes of PBS has not been explored. To bridge this gap, this study examines the moderating role of analytics capability on the link between PBS, green dynamic capability, and green innovation.

This paper combines insights from the DCT and OIPT to develop a moderated mediation model of how firms translate PBS into green innovation. By doing so, the paper advances knowledge on “how” and “when” PBS results in green innovation. Previous research has only focused on “how,” by testing mediation models without considering the boundary conditions therein (e.g., Wang & Yang, 2023; Xue & Wang, 2023). Also, this research introduces green dynamic capability as a mechanism through which PBS results in green innovation in response to calls for more research on how PBS relates to green innovation (Wang et al., 2019). Further, the research illuminates the role of analytics capability in this research stream, as firms need such capability to develop actionable intelligence from the chunk of data provided by PBS (Singh & Singh, 2019). Finally, the research uses data from a developing country context where due to limited access to structured environmental data, knowledge gaps, and low technical expertise (Nguyen & Adomako, 2022), firms need to engage in more PBS to develop green innovations.

2 | THEORY AND HYPOTHESIS

2.1 | Dynamic capabilities theory

The DCT originates partly from criticism and further development of the resource-based view (RBV) of the firm (Beske, 2012) to explain how firms achieve competitive advantage in a changing, rather than static environment (Teece, 2014). Dynamic capability is a firm's ability to flexibly use and adapt its skills and resources, both from within the company and from the outside, to address rapidly changing environments (Teece et al., 1997). With dynamic capabilities, firms can build, renew, reconfigure, and combine resources and assets to innovate (Teece, 2014). Three broad clusters of dynamic capabilities: sensing opportunities, seizing opportunities, and transforming business models provide the capability to purposefully create, modify, and extend a firm's resource base (Helfat et al., 2007). Prior studies suggest that dynamic capabilities are a precursor to green innovation (Nur & Kabadurmus, 2020). Green innovation requires that firms

merge knowledge from heterogeneous sources and reconfigure external knowledge. This suggests that firms require dynamic capabilities to support green innovation development. Also, sustainable environmental practices are non-static (Achi et al., 2022), which requires that firms build proactive pro-environmental capabilities. Therefore, this study draws on the DCT to argue that PBS enhances green innovation through the development of green dynamic capabilities. Green dynamic capabilities enable firms to seize and reconfigure the knowledge obtained from PBS, to build the required capabilities for green innovations (Yousaf, 2021).

2.2 | Organizational information processing theory

OIPT suggests that organizations should be designed to reduce uncertainty (Galbraith, 1974). This uncertainty is caused by the disparity between the amount of information that is required to perform a task and the amount of information that an organization already possesses (Haußmann et al., 2012). Firms can overcome uncertainties either by reducing their information processing needs or increasing their information processing capacities (Yu et al., 2019). Thus, high environmental uncertainty increases the information processing needs of the firm, which also increases the need for higher information processing capacity (Yu et al., 2019). Firms that can obtain a “fit” between their information processing capacity and their information processing needs can attain desirable performance (Pu & Qiao, 2023). Information processing capacity refers to the organization's capability that support gathering, interpreting, and summarizing information to support decision-making (Zhu et al., 2018). This study draws on the OIPT to argue that firms, when they engage in PBS acquire novel and unstructured data that requires significant data processing capabilities. Output from PBS is large amounts of unstructured data that require high-level analytical capabilities to derive meaningful insights (Gu et al., 2021). The unstructured nature of such data creates uncertainties and risks that can be mitigated by improving information processing capabilities (Galbraith, 1974). Therefore, firms require analytics capability—which enables them to process large amounts of data quickly- to derive meaningful insights for better decision-making (Zhu et al., 2018). Thus, while firms with high analytics capability can efficiently process data from PBS to reduce the associated uncertainty, firms with low analytics capability may add to struggle to gain meaningful insights from the data, leading to uncertainty in decision-making. Drawing on the OIPT, this study examines the moderating role of analytics capability on the link between PBS, green dynamic capability, and green innovation.

2.3 | PBS, green dynamic capabilities, and green innovation

Green dynamic capability refers to a firm's ability to innovate and bring positive change in response to changing environmental conditions (Aftab et al., 2024). Firms engage in PBS because internal

knowledge sources are often insufficient to generate the required capabilities for green innovation (Yang et al., 2021). PBS promotes the development of green dynamic capabilities by making available valuable external knowledge sources that support the development of internal capabilities. By engaging in PBS, firms acquire novel knowledge resources required for green innovations, which improve their green dynamic capability. Further, PBS nurtures the development of green dynamic capabilities as it helps to create collaborative networks, strengthens internal and external knowledge flows, and enhances the firm's ability to obtain environmental insights. Empirical evidence suggests that social networks and knowledge-sharing mechanisms, all of which are manifestations of PBS, enforce green dynamic capabilities. (Bresciani et al., 2023; Nassani et al., 2022). PBS provides essential knowledge from the external environment that enhances an organization's capacity to utilize resources to acquire new skills to support green innovations (Yousaf, 2021). Moreover, PBS sharpens the organization's ability to sense, seize, and reconfigure external environmental information. PBS also generates capabilities to produce, apply, and combine novel knowledge (Stefan & Bengtsson, 2017). When firms engage customers, suppliers, research institutions and other stakeholders, they obtain relevant information that helps to develop new skills that are required to support green innovation development (Nassani et al., 2022). Since green dynamic capabilities stress integrating, producing, and regenerating internal and external resources to facilitate environmental actions (Aftab et al., 2024), PBS provides external knowledge resources that facilitate the development of green dynamic capabilities. On this premise, it is hypothesized that:

H1a. PBS is positively related to green dynamic capability.

Building dynamic capabilities is a precursor to proactive innovation (Nur & Kabadurmus, 2020). Green dynamic capabilities assist organizations to learn modern green innovation techniques (Aftab et al., 2024). Organizations with strong green dynamic capability possess the requisite skills and abilities to effectively develop novel environmentally beneficial products, services, and technologies (Dangelico et al., 2017; Li et al., 2023; Singh et al., 2022). When companies with strong green dynamic capability identify opportunities for green innovation, they can fully mobilize and allocate resources such as capital, manpower, and knowledge required for green innovation (Yuan & Cao, 2022). Also, green dynamic firms have a strong desire to adopt new innovative and sustainable solutions for their customers, increasing green innovation (Qiu et al., 2020). Firms use green dynamic capabilities to create new environmentally friendly products by combining existing information with the acquisition of new knowledge (Bresciani et al., 2023). Green dynamic capability thus helps firms to integrate external resources with internal ones to create usable resources that support green innovations (Dangelico et al., 2017). Companies with strong green dynamic capability have stronger resource reorganization capability, which enables them to quickly organize resources to change environmental demands. Firms deploy green dynamic capability to reconfigure cross-functional teams to form new resource

combinations that can effectively implement green innovations (Yuan & Cao, 2022). Additionally, several empirical studies have found that green dynamic capability is positively related to green innovation (Aftab et al., 2024; Bresciani et al., 2023; Liboni et al., 2023; Nassani et al., 2022). Based on this premise, this study hypothesizes that:

H1b. Green dynamic capability is positively related to green innovation.

While PBS provides crucial knowledge resources for developing green innovations, it can also result in information overload that derails green innovation (Shi et al., 2019). Therefore, firms must orchestrate and leverage the information obtained through PBS to enhance green innovation (Yang et al., 2021). Consequently, some researchers have highlighted that mediating variables may be critical in evaluating the PBS and green innovation nexus (e.g., Wang & Yang, 2023; Xue & Wang, 2023). Knowledge acquired through boundary-spanning search does not provide direct benefit for firms, and the firm's current knowledge and capability determine how much external knowledge can be utilized by the firm (Wang et al., 2019). The literature considers building dynamic capabilities a precursor to proactive innovation (Nur & Kabadurmus, 2020). Building on this literature stream, this study positions green dynamic capability as an intervening mechanism that enables firms to exploit PBS to create green innovations (Qiu et al., 2020). PBS generates capabilities to produce, apply, and combine novel knowledge, thereby boosting innovation performance (Stefan & Bengtsson, 2017). PBS sharpens the firm's (green) dynamic capability for sensing, seizing, and reconfiguring internal and external knowledge. The green dynamic capability then enables firms to exploit the external search knowledge to create green innovations (Qiu et al., 2020). When firms search the external environment for knowledge resources, it complements their internal capabilities to build new resources that support green innovation development. Therefore, green dynamic capability bridges the relationship between PBS and green innovation. Based on this premise, this study argues that:

H1c. Green dynamic capabilities mediate the relationship between proactive boundary-spanning search and green innovation.

2.4 | The moderating role of analytics capability

Analytics capability is the ability to “process large volumes and varieties of data with the velocity required to gain relevant insights” (Liu & Wei, 2022). PBS could result in large unstructured data that requires analytics capability to make sense of the data. Analytics capability helps to process large volumes, high velocity, and several data varieties to extract useful insights (Wamba et al., 2018) to support green innovation. Therefore, analytics capability helps firms to navigate the information processing needs presented by PBS (Galbraith, 1974) to reduce the uncertainty and risks posed by making decisions with novel and unstructured data. Previous studies suggest that analytics

provide new methods to process and analyze valuable information that can assist decision-making (Zameer et al., 2020) in green innovation development. Firms with analytics capability are in a better position to assemble, integrate and deploy knowledge from their PBS to create green innovations (Wang et al., 2020). When analytics capability is high, firms can rapidly and more effectively analyze the data from their external search (Zameer et al., 2020), reveal meaningful directions, and build the needed competencies to support green innovations. High analytics capability helps firms to sense, seize, and disseminate useful environmental information for managerial decision-making. Analytics capability also improves information processing capacity and this reduces the uncertainty and risks associated with green innovations (Wamba et al., 2018). Therefore, analytics capability enhances firms' ability to turn PBS results into useful to enhance the firms' green dynamic capabilities. Data analytics affects firms' anticipation and improvisation to support the development of green dynamic capabilities, which in turn enhances green innovations. Conversely, firms with low analytics capability have limited capacity to transform external search results into capabilities that can support green innovation (Wu & Huang, 2018). These firms can easily be overwhelmed with excessive information that can derail green process innovation. Firms with low analytics capability are likely to interpret the data from PBS slowly (Liu & Wei, 2022), and miss insights that can support the development of capabilities for green innovation. Overall, analytics capability increases a firm's information processing capacity to reduce uncertainty and risks (Munir et al., 2022) PBS for green innovations.

Based on this premise, this study argues that:

H2. Analytics capability positively moderates the indirect relationship between proactive boundary-spanning search and green innovation via green dynamic capabilities.

The proposed model is presented in Figure 1.

3 | METHODOLOGY

3.1 | Survey and data collection

The study's theoretical model is tested with firm-level survey data collected from manufacturing firms in Ghana. The sampled firms operate

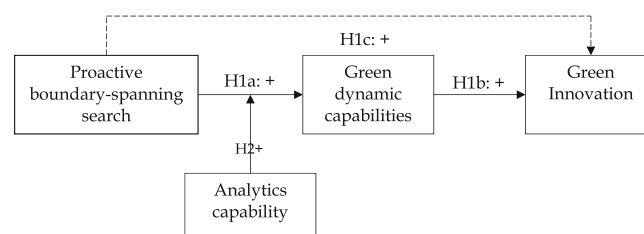


FIGURE 1 Research model.

in the two most industrialized regions in the country—Ashanti and Greater Accra. Ghana has been on an upward development path and is considered one of the fastest-growing economies in Sub-Saharan Africa (World Bank Group, 2018). Recent governments have sought to increase the industrial base of the economy by pursuing industrialization policies such as the “one-district-one-factory” agenda, together with other incentives that seek to attract foreign direct investment in the manufacturing sector. This has increased the scale and intensity of manufacturing operations, which raises concerns about the environmental impact of these operations. Also, there is an increased focus on environmental issues in the country following numerous reports on illegal mining and the associated environmental menace (Osei & Atta-ankomah, 2020).

The items used to measure the constructs were all adapted from previous research. Before administering the survey, a pilot test was conducted using 27 graduate students enrolled in a supply chain management program and holding at least middle management roles in the industry. The results of an exploratory factor analysis show that all items loaded on their respective factors. The final survey was administered to 307 firms identified through an online business directory. All questionnaires were delivered in person by field agents to the companies. An introductory letter accompanied each questionnaire to explain the purpose of the study and also to assure participants of anonymity and confidentiality of their responses.

Although the target respondents were environmental/innovation/production and/or supply chain managers, the guidelines encouraged collaboration among top managers in responding to the survey, as a way of reducing common method bias. After several rounds of reminders (using phone calls to contact persons) and on-site visits by field agents, 204 questionnaires were returned. Compared to previous studies that used a survey method, the response rate was high because field agents visited the companies directly and, in most cases, got to speak with the respondents before leaving the questionnaire with them. Through an initial check, 4 of the returned were considered unusable and dropped from the dataset. This left 200 usable questionnaires. The characteristics of the sample firms are presented in Table 1.

3.2 | Measures

PBS was measured using five items adapted from (Wang & Yang, 2023). The items are: “Our firm has many channels to search new domain knowledge ahead of competitors,” “Our firm can search new domain knowledge about R&D, production, and marketing ahead of competitors,” “Our firm can search new domain knowledge about technology and management ahead of competitors,” “Our firm can search more new knowledge ahead of competitors,” and “our firm can conduct in-depth search and obtain knowledge of emerging technologies ahead of competitors.”

Three items from Yousaf, (2021) relating to the firm's ability to translate external data into useful green information were used

to measure green dynamic capabilities. The items are: “The company has the ability to assimilate, learn, generate, combine, share, transform, and apply new green knowledge,” “The company has the ability to successfully integrate and manage specialized green knowledge within the company,” and “The company has effective routines to identify and develop new green knowledge.”

Analytics capability is measured with five items from Liu & Wei, (2022) examining the extent to which firms use data management and analytical techniques. The items are “We use advanced analytical techniques (e.g., simulation, optimization, and regression) to improve decision-making,” “We easily combine and integrate information from many data sources for use in our decision-making,” “We routinely use data visualization techniques (e.g. dashboards) to assist users or decision = makers in understanding complex information,” “Our dashboards give us the ability to decompose information to help root cause analysis and continuous improvement” and “We deploy dashboard applications/information to our managers' communication devices (e.g., smartphones and computers).”

Green innovation is measured using three items that tap into the domain of green product, process, and managerial innovation. The items are: “Our company has introduced new products that seek to reduce environmental impacts,” “our company has redesigned its production process to be environmentally friendly” and “Our management regularly considers new ways to improve our operations to improve our environmental performance.” All the items were measured on a seven-point scale anchored from strongly disagree (1) to strongly agree (7).

TABLE 1 Company characteristics.

Characteristics	Frequency	Percent
<i>Manufacturing industry type</i>		
Industrial machinery	12	6.0
Chemicals	14	7.0
Plastics and rubber	10	5.0
Food, beverages and drinks	96	48.0
Metals and metalworking	12	6.0
Pharmaceuticals	10	5.0
Paper and packaging materials	3	1.5
Engineering and construction	19	9.5
Textiles and clothing	6	3.0
Others	18	9.0
Total	200	100.0
<i>Ownership type</i>		
Local private company	173	86.5
Subsidiary of a foreign multinational	5	2.5
Local public company	22	11.0
Total	200	100.0
<i>Other characteristics</i>		
Number of employees (4–352)	32.45	43.582
Firm age (6–27 years)	13.6150	3.80647

3.3 | Bias assessment

To assess potential non-response bias, a comparison between early (first 15% of responses received) and late responses (last 15% of responses received) was conducted. The independent sample *t* test was conducted using the firm characteristics and the substantive variables revealed no significant statistical difference between the two groups. This suggests that non-response bias is not a serious concern for this study.

Common method bias potentially affects the results of cross-sectional studies that use single respondents (Podsakoff et al., 2012). This study used several procedural (ex-ante) techniques to minimize common method bias. First, all respondents were assured that their responses would be treated with confidentiality and that the data collected was solely for academic purposes. Additionally, the questionnaire did not collect any respondent or firm-specific data that could influence the respondent to provide socially desirable responses. Further, the items in the questionnaire were randomized so the respondents cannot identify any theoretical logic between the variables. Statistical (post-ante) procedures have been used after the data collection to evaluate the potential existence of common method bias. First, Harman's one-factor

test using exploratory factor analysis shows that the first component explained only 25.7% of the total variance, which is under the 50% threshold. Further, following the recommendations of Kock, (2015) a full collinearity test approach was used to examine the CMB by examining Variance Inflation Factors (VIF). All the values were under the 3.3 threshold, demonstrating that the CMB was not a severe concern in the dataset (Aftab et al., 2024; Kock, 2015).

4 | RESULTS

4.1 | Measurement model

Partial least squares structural equation modeling (PLS-SEM) was used to examine the measurement model. PLS-SEM has been chosen for several studies in environmental management due to its ability to produce consistence estimates without imposing strict rules of data normality and sample size (Aftab et al., 2024). PLS-SEM is also preferred for estimating complex models such as moderated mediation models (Hair et al., 2021). Model fit examination is still an emerging area yet to achieve full resolution in PLS-SEM modeling (Hair et al., 2019; Hair et al., 2020). However, SMART PLS 3 produces estimates of the standardized root mean residual (SRMR) and Normed Fit Index (NFI) which provides an initial assessment of model fit. The SRMR and NFI of this study's model are 0.068 and 0.912 respectively, which compare favorably with the widely accepted thresholds (SRMR ≤ 0.08 and NFI ≥ 0.90) to indicate model fitness (Hair et al., 2016).

Table 2 presents the factor loadings and tests of convergent and internal consistency. All factor loadings, composite reliability, and Cronbach's alpha values are above the recommended value of 0.7. The Average variance extracted value is above 0.5, indicating convergent validity.

Table 3 presents the test of discriminant validity using the Heterotrait-Monotrait (HTMT) ratio. Following conventional rules, the highest HTMT is below 0.8 and this suggests that discriminant validity is achieved. To further demonstrate discriminant validity, the square root of the AVEs was compared with the correlations between the variables. Table 4 indicates that the square roots of AVEs (which are presented in bold and italicized on the diagonal) are

TABLE 2 Confirmatory factor analysis, reliability, and validity tests.

Construct	Loading	VIF
<i>Analytics capability</i> (CR = 0.935, AVE = 0.741, CA = 0.913)		
AC1	0.836	2.219
AC2	0.879	2.935
AC3	0.846	2.705
AC4	0.887	2.351
AC5	0.854	2.665
<i>Proactive boundary-spanning search</i> (CR = 0.920, AVE = 0.696, CA = 0.890)		
BSS1	0.767	1.787
BSS2	0.797	1.942
BSS3	0.864	2.712
BSS4	0.879	2.046
BSS5	0.859	2.506
<i>Green dynamic capability</i> (CR = 0.873, AVE = 0.679, CA = 0.779)		
GDC1	0.760	1.416
GDC2	0.894	2.276
GDC3	0.845	1.902
<i>Green innovation</i> (CR = 0.878, AVE = 0.708, CA = 0.792)		
GI1	0.721	1.375
GI2	0.899	2.157
GI3	0.891	2.131

TABLE 3 Heterotrait-monotrait ratio (HTMT).

	AC	BSS	GDC	GI
AC				
PBS	0.588			
GDC	0.435	0.627		
GI	0.130	0.250	0.687	

Abbreviations: AC, analytics capability; GDC, green dynamic capability; GI, green innovation; PBS, proactive boundary-spanning search.

TABLE 4 Correlation matrix.

	1	2	3	4	5	6
1. Analytics capability	0.860					
2. PBS	0.526**	0.834				
3. Green dynamic capability	0.367**	0.524**	0.834			
4. Green innovation	0.108	0.208**	0.541**	0.841		
5. Firm size	0.233**	0.268**	0.215**	0.276**	n/a	
6. R&D	0.325**	0.376**	0.291**	0.197**	0.275**	n/a
Mean	3.84	3.77	4.31	4.53	32.4	8.86
Standard deviation	1.34	1.25	1.24	1.26	43.5	5.59

**Correlation is significant at the 0.01 level (2-tailed).

TABLE 5 Hypothesis test results.

	β	SE	t	p
<i>Direct effects</i>				
AC $\rightarrow$ GDC	0.163	0.075	2.171	0.030
BSS $\rightarrow$ GDC (H1a)	0.433	0.076	5.682	0.000
GDC $\rightarrow$ GI (H1b)	0.556	0.051	10.927	0.000
BSS $\rightarrow$ GI	-0.083	0.066	1.250	0.212
<i>Mediating effect</i>				
BSS $\rightarrow$ GDC $\rightarrow$ GI (H1c)	0.240	0.045	5.282	0.000
<i>Interaction effects</i>				
BSS $\times$ AC $\rightarrow$ GDC	0.284	0.075	3.783	0.000
BSS $\times$ AC $\rightarrow$ GDC $\rightarrow$ GI (H2)	0.158	0.049	3.204	0.001

greater than the highest correlations between the variables, further providing support for discriminant validity among the constructs.

4.2 | Structural model

The structural model is tested using PLS-SEM and conditional process analysis in Hayes PROCESS. The characteristics of the structural PLS-SEM model are evaluated based on the coefficient of determination (R^2) and the effect size (f^2) of the predictors in the model (Benitez et al., 2020). First, the R^2 -values, which indicate the proportion of variance explained in the endogenous variables are 33.9% for green dynamic capability and 30.9% for green innovation. Second, the f^2 values for the main paths in the model are; 0.201 for BPS $\rightarrow$ green dynamic capability, 0.446 for green dynamic capability $\rightarrow$ green innovation and 0.190 for the interaction term $\rightarrow$ green dynamic capability. These f^2 values indicate moderate ($0.15 \leq f^2 \leq 0.35$) and substantial effect sizes ($f^2 \geq 0.35$), according to (Cohen, 1988).

Table 5 presents the results of the hypothesis tests. First, the results show that PBS is positively related to green dynamic capability ($\beta = 0.433$, $t = 5.682$). This supports hypothesis H1a. The results also show that green dynamic capability is positively related to green

TABLE 6 Conditional effects of PBS on green dynamic capabilities at different levels of analytics capability.

AC	Effect	LLCI	ULCI
Low (-1 SD)	0.163	-0.030	0.356
Moderate (mean)	0.425	0.285	0.564
High ($+1$ SD)	0.687	0.490	0.884

innovation ($\beta = 0.556$, $t = 10.927$), supporting hypothesis H1b. The results further show that green dynamic capability mediates the relationship between PBS and green innovation. The indirect effect is significant at 5% ($\beta = 0.240$, $t = 5.282$), supporting hypothesis H1c. The results also show that the direct effect between PBS and green innovation is not significant ($\beta = -0.083$, $t = 1.250$), thus indicating that the mediation is full. The results further show that the relationship between PBS and green dynamic capability is moderated by analytics capability ($\beta = 0.284$, $t = 3.783$). Further exploration of this relationship using the bootstrapping approach in Hayes PROCESS, as presented in Table 6 shows that at higher levels of analytics capability, PBS is positively related to green dynamic capability as the 95% bootstrapping confidence interval does not include a zero (LLCI = 0.490, ULCI = 0.884). At lower levels of analytics capability, the relationship between PBS and green dynamic capability is not significant, as the 95% bootstrapping confidence interval contains a zero (LLCI = -0.030, ULCI = 0.356).

The results from Table 5 further show that as proposed in hypothesis 2, the indirect relationship between PBS and green innovation via green dynamic capability is moderated by analytics capability. This is supported by the results of the conditional PROCESS analysis in Hayes PROCESS model presented in Table 7. The index of moderated mediation is significant (Index = 0.113, LLCI = 0.045, ULCI = 0.190), indicating that the moderated mediation model is significant (Hayes & Rockwood, 2020). At higher levels of analytics capability, firms' PBS translates into green innovation via green dynamic capability ($\beta = 0.399$, LLCI = 0.251, ULCI = 0.557). Conversely, when analytics capability is low, PBS does not translate into green innovation via green dynamic capabilities ($\beta = 0.095$, LLCI = -0.038, ULCI = 0.238).

Path	Level of AC	Indirect effect	LLCI	ULCI
BSS → GDC → GI	Low	0.095	−0.038	0.238
	Medium	0.247	0.147	0.359
	High	0.399	0.251	0.557
	Index	SE	LLCI	ULCI
<i>Index of moderated mediation</i>	0.113	0.038	0.045	0.190

Note: $n = 200$, Bootstrap sample = 5000.

Abbreviations: AC, analytics capability; BSS, boundary spanning search; GDC, green dynamic capability; GI, green innovation; LLCI, Lower limit confidence interval; ULCI, Upper limit confidence interval.

TABLE 7 Conditional indirect effect of PBS on green innovation via green dynamic capability at different levels of analytics capability.

5 | DISCUSSION

This paper has examined the relationship between firms' PBS and green innovation of manufacturing firms in Ghana. The results extend the scholarly discussion in this area by providing a nuanced view of how firms' analytics capability conditions their ability to transform PBS into dynamic capabilities that create green innovations. The results suggest that green dynamic capabilities mediate the relationship between PBS and green innovation, and this indirect relationship is positively moderated by analytics capability. The study has several theoretical implications, which have been explained below.

5.1 | Theoretical implications

First, this study provides a comprehensive view of how firms' PBS search translates into green innovations as research in this area is still nascent (Wang & Yang, 2023). The study contributes to research in this area by considering green dynamic capability as a mechanism and analytics capability as a condition that helps firms develop green innovations from their PBS. Therefore, this paper extends the discussions in Wang et al. (2019) and Wang and Yang (2023) on the link between PBS and green innovations. This paper corroborates the position of existing studies that although PBS is crucial for developing green innovations, it may not directly influence green innovations (Wang et al., 2019). The results confirm a positive relationship between PBS and green dynamic capabilities, as well as a positive relationship between green dynamic capabilities and green innovation. These findings set the study up for examining the proposed indirect relationship. Overall, the model provides a theoretical explanation for the mechanism through which firms can channel their PBS search into successful green innovations.

Second, this study shows that green dynamic capability mediates the relationship between PBS and green innovations. Previous studies identified resilience and resource orchestration capabilities in the same link (Wang et al., 2019; Xue & Wang, 2023). This finding extends the ongoing discussions by introducing green dynamic capability in the link between PBS and green innovation. The results suggest that PBS is translated into green dynamic capabilities that enable the development of successful innovations. Several studies consider green dynamic capabilities as a prerequisite for green innovations (Bresciani

et al., 2023; Yousaf, 2021; Yuan & Cao, 2022). This study advances this line of research by presenting a comprehensive model that explains how PBS helps to develop green dynamic capabilities, which then supports green innovation development. While studies have shown that green dynamic capabilities positively affect green innovations (Hong et al., 2018; Qiu et al., 2020; Singh et al., 2022), little attention is paid to what firm-level strategies or actions support the development of green dynamic capabilities. Further, this research supports extant studies that approach green innovation from a DCT perspective to consider how dynamic capabilities help firms to develop green innovation in response to stakeholder environmental demands.

Third, the research model explains how the information processing capabilities of the organization help to translate PBS into green dynamic capabilities to support successful innovations. Although the potential for information overload is recognized in previous studies (Wang & Yang, 2023), there had been little focus on how firms' ability to analyze and make sense of data—analytics capability—enables them to make more meaning of their external search, and further translate to capabilities for green innovations. Drawing on the OIPT, the results suggest that analytics capability increases the information processing power of the firm enables them to reduce task uncertainty (Haußmann et al., 2012; Yu et al., 2019). This finding enriches scholarly understanding of what firm capabilities are required to increase the value obtained from PBS for environmental knowledge. Analytics capability improves the information processing capacity of organizations, which helps to reduce the uncertainty and risks associated with innovations (Wamba et al., 2018), especially in developing countries. Technological advancements have increased the amount of data available to firms in the external environment, and firms require significant capabilities to analyze and make sense of that data (Aker et al., 2016). Beyond this, the study also contributes to improving the limited understanding of how analytics support operations from a supply chain perspective (Munir et al., 2022).

Fourth, this research makes a timely contribution to the literature on the PBS for environmental knowledge using data from a developing economy in Sub-Saharan Africa. There is mounting environmental pressure on African nations amid fears of missing their sustainable development goals (Aminu et al., 2023). Stakeholder interest in environmental issues is on a steady rise, and firms have begun to search for external knowledge to support their green innovations (Appiah, 2023). Given the relatively low level of

technological development, low experience and lack of existing data, PBS is likely to return large chunks of unstructured data. This necessitates that firms need to develop capabilities to derive useful information from huge unstructured data from the external environment. This study offers a comprehensive view of how firms in these contexts can develop successful innovations from their PBS for environmental knowledge.

5.2 | Managerial implications

The results provide the following managerial implications. First, the results confirm that PBS promotes green dynamic capability and subsequently green innovation. Therefore, managers should actively integrate PBS into their business operations to obtain novel environmental knowledge. Firms can enhance their PBS by first creating an environment that considers PBS a value driver instead of a cost driver. With such orientation, managers should be intentional about creating collaborative links with customers, suppliers, research institutions and other environmental stakeholders that are potential sources of innovative knowledge. Specifically, firms should have laid down procedures for networking to obtain environmental ideas from external stakeholders—universities, government agencies, research agencies, non-governmental organizations, customers, suppliers, competitors and so forth—to facilitate green innovations. Being part of an organizational network facilitates access to resources that can impact innovation performance (Sebaka & Zhao, 2022). Managers can also enhance their PBS by investing in technological resources that assist in identifying, collecting, and transmitting relevant knowledge from the external environment, such as blockchain technology and the Internet of Things (IoT).

Second, the results show that green dynamic capabilities mediate the relationship between PBS and green innovation. Therefore, firms should take steps to enhance their green dynamic capabilities. Managers can do so by creating a dedicated and competent environmental protection department in the firm that coordinates the efforts of research and development, production, marketing, engineering, supply chain, and other departments on environmental management. Such cross-functional collaborations can help to assimilate the knowledge from PBS to create capabilities that support green innovations. Firms should make a concerted effort to build positive working relationships with the government, suppliers, and partners and put efforts into obtaining and sharing resources required to improve their green dynamic capabilities. Additionally, managers should develop specific learning outcomes and key performance indicators that will be used to track how the firm has learned from its environmental knowledge, and what new skills have been acquired to help its environmental management practices.

Third, the results show that firms with high analytics capability are better poised to translate their PBS into green innovations. Such analytics capability enables firms to process the large chunk of information that results from the external search into meaningful data, while also allowing firms to avoid information overload that can derail

their green innovation efforts. Therefore, firms should seek to improve their analytical capabilities by investing in data collecting, processing, and summarizing technologies, as well as training staff on data analytics. At the strategic level, firms should have a pro-analytics culture that supports data-based decision-making. Firms with a pro-analytics culture will constantly scan the market for new tools and techniques to improve their ability to derive insights from environmental data. Firms should also build networks with institutions that research and develop statistical models and other analytics procedures so they can obtain early information about new developments in the field that can improve the firm's analytics capabilities.

6 | CONCLUSION AND LIMITATIONS

Following the rise in stakeholder concerns about corporate environmentalism, many firms are seeking to develop new products and processes that concurrently meet environmental and economic goals. Because innovation seldom occurs in a single organization, firms are required to draw on external knowledge to support their green development efforts. This study is one of the earliest to examine how firms can translate their PBS search for green ideas into capabilities that can support successful green innovations.

Although this research makes a significant contribution to environmental sustainability literature, some limitations point to future research directions. Firstly, this research uses data from Ghana, and although this context has similar characteristics with other developing African nations, the findings may not apply to other countries or continents. Future studies can test the applicability of this research model to different settings. Second, the study investigated the mediating role of green dynamic capability and the moderating role of analytics capability in the baseline relationship. Other variables such as absorptive capacity, green value co-creation, and proactive behavior can be studied in the future to see how they affect the link between PBS and green innovation.

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